

Implement multimodality imaging integration (TEE, cardiac CT, [18F]-FDG PET/CT) into the heart valve prosthesis monitoring protocol

Executive Technology Brief

Domain: fluidics_hydraulic

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented recurring problem 'insufficient_device_problem_information' for heart valve prosthesis: 18 MAUDE reports across 4 distinct devices (20171025 to 20250513); root-cause evidence from 0 recall records and 10 event narratives (custody attached in the campaign report). MAUDE limitations: incidence UNKNOWN, causation UNVERIFIED (Art. XXI.5). Existing solutions fail because: Why existing solutions fail, per retrieved evidence: 10 MAUDE event narratives from this mechanism cluster (mdr_text, custody attached). Statements are the providers' own words, custody-attached; the aggregation across records is AI_INFERENCE (bounded by the r Unresolved gap: Across the 6 relevant retrieved attempts (a bounded, non-exhaustive set), none measurably addresses: conditions, devices, events, failure, function, injury, insufficient, intervention, introducing, maintain, matching, modes, must, problem, producing, recurring

Mechanism: Diagnostic challenges due to acoustic shadowing on echocardiography

Architecture: PROPOSED architecture (no physical system exists): flow path / lumen architecture; pressure regulation element; sensing / feedback element (if closed-loop); termination interfaces (patient / reservoir) — realizing: Implement multimodality imaging integration (TEE, cardiac CT, [18F]-FDG PET/CT) into the heart valve prosthesis monitoring protocol

Governing Model: Fluid mechanics / hydraulics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

FLUID-001: Hagen-Poiseuille (laminar pipe flow). Expression: $Q = (\pi * r^4 * \Delta P) / (8 * \mu * L)$.

Applicability to this invention: CONDITIONAL; holds under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow. Assumptions: CSF approximated as Newtonian (valid at shunt flow rates); no wall compliance effects. Engaged variables: NOT ESTABLISHED.

FLUID-002: Reynolds number. Expression: $Re = (\rho * v * D) / \mu$. Applicability to this invention:

CONDITIONAL; holds under: circular section; single-phase flow. Assumptions: transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb. Engaged variables: NOT ESTABLISHED.

FLUID-003: Orifice flow (sharp-edged restriction). Expression: $Q = C_d * A * \sqrt{2 * \Delta P / \rho}$.

Applicability to this invention: CONDITIONAL; holds under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured. Assumptions: $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement. Engaged variables: NOT ESTABLISHED.

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.